

CKMS replication and a complexity sweep of the maximal Sharpe ratio

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1 Setup

CKMS (version of 3 May 2026) bound the population maximal squared Sharpe ratio of a span of characteristic managed portfolios and report an annualized figure near 7.4 for United States equities. I rebuilt their equity pipeline on the supplied JKP panel. The object and the reported bounds are

$$R_*^2 = \mu' \Sigma^{-1} \mu, \quad H_1^L \leq R_*^2 \leq H_1^U, \quad \text{reported as } \sqrt{12 H}.$$

The managed portfolio returns and the nonlinear features are

$$F_{t+1} = \frac{1}{N_t} S_t' r_{t+1}, \quad S_t = \text{ReLU}\left(\frac{1}{2} \sqrt{2/d} X_t \Omega'\right), \quad \Omega_{pj} \sim N(0, 1) \text{ drawn once.}$$

X_t holds each characteristic ranked cross sectionally into $[-0.5, 0.5]$ with missing entries set to zero. Table 1 lists the specification and its source.

Table 1. Specification. Items marked CKMS follow the paper.

Item	Value	Source
Sample	756 months, Jan 1963 to Dec 2025	CKMS
Size groups	mega, large, small, micro	CKMS, NYSE 80/50/20/1 pct
Characteristics kept	121	< 30% missing rule, CKMS report 132
Missing return rows	dropped, no imputation	this build
Random features	$P = 5000$ ReLU, He scale, fixed Ω	CKMS eq. (49), (51)
Estimator	$J = 1$ bounds, delta method band	CKMS App. C, H
Ridge z	10^{-4} nominal, see text	CKMS
Rolling window	$T = 240$ months	CKMS
Eligible stock months	2,635,914	CRSP id, non nano, chars present
Used, return seen next month	2,627,813	this build
Monthly stock count, all	737 / 3,627 / 6,271 (min, med, max)	this build

The characteristic count is 121 against the 132 CKMS report. CKMS keep a characteristic when it is under 30 percent missing, and applying that same rule to this JKP data vintage and stock universe keeps 121. The gap is most likely a data vintage or an undocumented preprocessing difference, and I treat it as a minor discrepancy.

The larger open question is the ridge. The literal specification is $z = 10^{-4}$, and on this build's managed return covariance no single z reproduces the full sample effective shrinkage $\hat{Z}'_*$ that

CKMS report near 1.8 while keeping every 240 month window well posed. I therefore carry two conventions through every result,

$$\text{fixed } z = 2.5 \times 10^{-7}, \quad \text{matched } z \text{ per estimate so that } \hat{Z}'_*(z) = 1.4,$$

fixed z being the smallest value that keeps every window well posed and 1.4 the highest $\hat{Z}'_*$ reachable under that constraint. They isolate the regularization from opposite sides. Fixed z holds the estimator identical everywhere and lets $\hat{Z}'_*$ rise with P ; matched $\hat{Z}'_*$ holds $\hat{Z}'_*$ constant and lets z fall as P rises. Neither is privileged, and every qualitative finding in Sections 3 and 4 is the same under both. A separate value $z = 3 \times 10^{-8}$ recovers the CKMS full sample figure but makes the rolling windows degenerate, so it appears once as a scale check.

2 Replication

Table 2. Full sample annualized maximal Sharpe ratio (L / U). Paper values are the Figure 4 boxes in CKMS. Scale check uses $z = 3 \times 10^{-8}$ and is a calibration diagnostic, fixed z uses $z = 2.5 \times 10^{-7}$.

Universe	Paper	Scale check	Fixed z
Mega	2.20 / 2.23	2.73 / 2.75	2.34 / 2.45
Large	3.52 / 3.56	3.72 / 3.74	3.20 / 3.33
Small	4.84 / 4.92	4.96 / 4.99	4.34 / 4.51
Micro	7.32 / 7.34	8.01 / 7.87	6.73 / 6.88
All	7.43 / 7.42	7.50 / 7.48	6.41 / 6.57

At the scale check the all stock pair is within about 1 percent of CKMS and four of five size groups are within about 10 percent with the ordering preserved, which is what the scale check is set up to do rather than evidence of a match. At fixed z the all stock linear estimate is 4.10 / 4.26, below the nonlinear 6.41 / 6.57, as in CKMS. The effective shrinkage $\hat{Z}'_*$ is about 1.3 at fixed z and about 1.7 at the scale check, against the CKMS value of 1.8, and the implied ridge ceiling $\sqrt{12L/\hat{Z}'_*}$ at fixed z is 5.8 against the CKMS value of 5.5. Figure 1 shows the rolling path.

3 Complexity sweep

The design holds the window fixed and expands complexity within it. Fix the window length at $T = 240$ months. At each of 517 monthly windows, from the one ending December 1982 to the one ending December 2025, estimate the bounds at 13 feature counts

$$P \in \{60, 100, 160, 250, 380, 560, 820, 1200, 1700, 2400, 3300, 4200, 5000\},$$

so $c = P/T$ runs from 0.25 to 20.8. The random matrix Ω is 5000×121 , drawn once, and a model with P features uses its first P rows, so the feature spans are nested, $\mathcal{S}_{P_1} \subset \mathcal{S}_{P_2}$ for $P_1 < P_2$, and a change in the estimate reflects added model capacity rather than a different random basis. Both ridge conventions are carried through. The result is a surface $SR_t(c)$, one value per window and feature count, shown in Figure 2. The plotted object is the $J = 1$ lower bound $SR_t^L(c) = \sqrt{12 H_{1,t}^L(c)}$, which is a genuine lower bound on the population $\sqrt{12 R_*^2}$. The paired upper bound and the midpoint of the two are within a median of about 1 percent for all stocks and 1 to 3 percent for the size groups, and every figure and summary below is unchanged in shape if the upper bound or the midpoint is plotted instead.

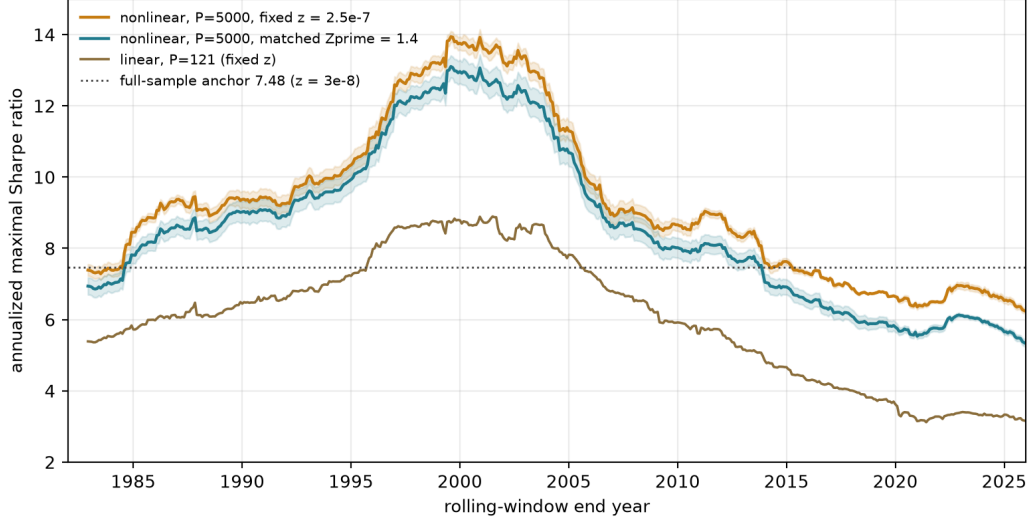


Figure 1. Rolling 240 month all stock estimate, $P = 5000$, both ridge conventions, with the linear estimate and the $z = 3 \times 10^{-8}$ scale check as the dotted line, labelled anchor in the panel. Bands span L to U .

A separate sweep adds nine feature counts between $P = 180$ and $P = 300$, so c steps through the $P = T$ region at 0.75, 0.83, 0.92, 0.96, 1.00, 1.04, 1.08, 1.17, 1.25. Under both ridge conventions the surface is smooth across this region, with no kink, spike or gap.

The estimate at the smallest model, $c = 0.25$, depends on which 60 of the 5000 features it uses. Drawing 40 random 60 feature subsets of the fixed Ω per window, the estimate has a cross subset coefficient of variation near 0.05, against about 0.02 at $c = 5$, and the first 60 columns used above sit within about 3 percent of the subset mean. The ratio $SR_t(c = 20.8)/SR_t(c = 0.25)$ therefore carries a feature choice uncertainty near ± 0.1 per window, which does not move its level or its lack of trend.

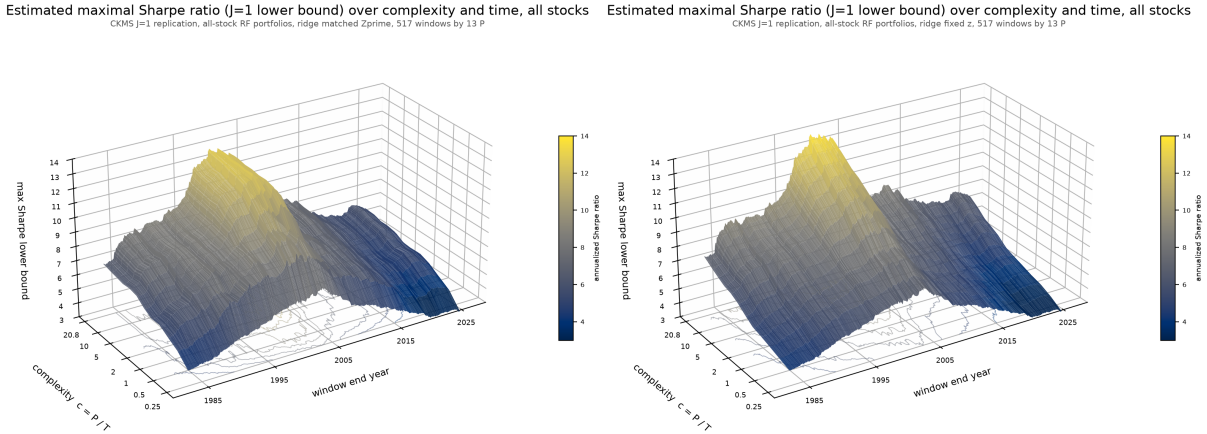


Figure 2. $SR_t^L(c)$, the $J = 1$ lower bound, over window end date and $c = P/T$, all stocks. Left matched $\hat{Z}'_*$, right fixed z . Each surface is 517 windows by 13 feature counts.

Because the spans are nested, the estimate is close to increasing in c by construction, so the informative content is not that Sharpe rises with c but how three things move across windows. The first is the level, read as the estimate at the largest model $SR_t(c = 20.8)$. The second is the shape, read from a singular value decomposition of the surface after removing the time mean at each c , which separates a common time factor from any change in the curve. The third is the

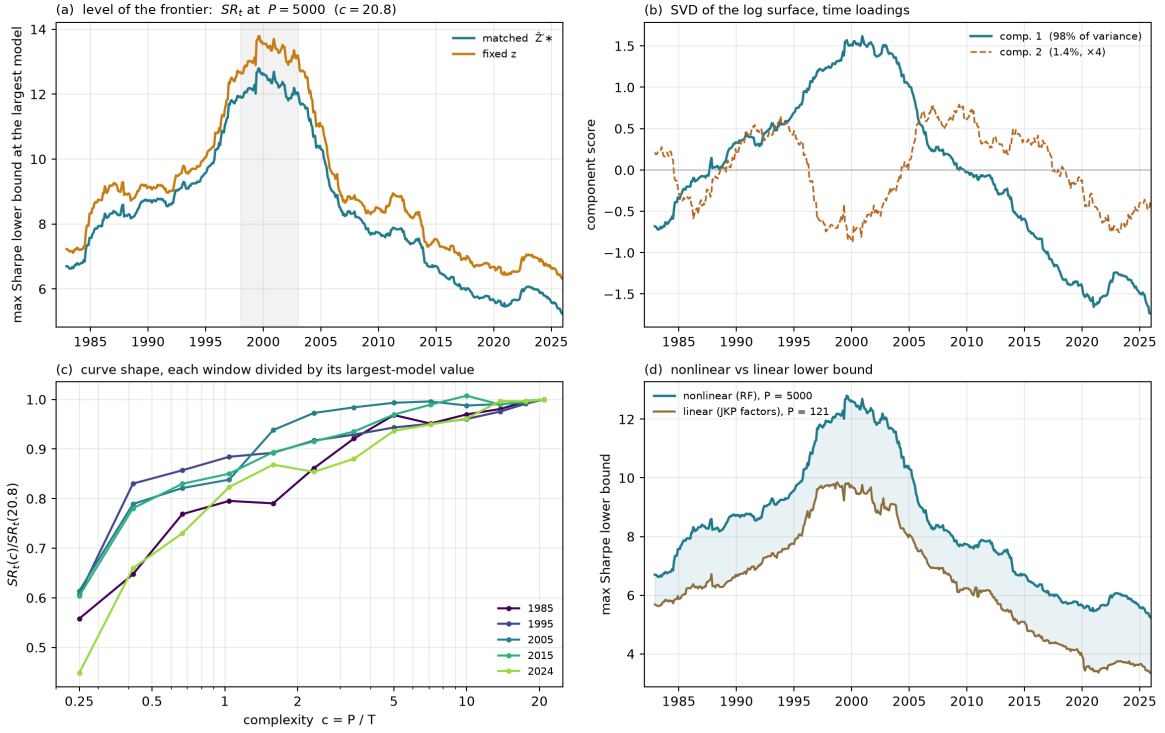


Figure 3. (a) the level $SR_t^L(c = 20.8)$ over time, both conventions. (b) the first two time components of a singular value decomposition of the log surface after removing the time mean at each c . (c) the curve for five windows, each divided by its own value at $c = 20.8$. (d) the largest nonlinear lower bound and the linear lower bound.

gain from added complexity, read model free as the ratio $SR_t(c = 20.8)/SR_t(c = 0.25)$ and the difference $SR_t(c = 20.8) - SR_t(c = 0.25)$, which show whether that gain moves in proportional or in absolute terms. Any trend across windows is read with a Newey-West standard error at 240 months, for the reason in the note below. Figure 3 and Table 3 report these.

The level rises from about 7 in the early 1980s to about 13 at the window ending in 1999, then falls back to about 5 by the end of the sample, the same on the upper bound. The singular value decomposition is taken on the log surface, and it puts 98 percent of the time variation on one component whose loading on c stays within 0.88 to 1.00 of its own maximum across the whole range, so to a close approximation

$$\log SR_t(c) \approx a_t + g(c), \quad \text{equivalently} \quad SR_t(c) \approx A_t G(c),$$

a fixed complexity shape G scaled by a time varying level A_t . The time variation is multiplicative in the level, not a change in the shape. Panel (c) shows the same thing, the five normalized curves nearly coincide. The ratio stays near 1.8 to 2.0 and the difference near 4 Sharpe points, neither with a clear trend, so over this sample a larger model buys a roughly constant advantage rather than a growing one. The largest nonlinear lower bound sits about 2 Sharpe points above the linear one throughout, as in CKMS, and that gap does not trend.

The 98 percent one component share could reflect the overlapping windows alone. A stationary check resamples the managed returns in circular blocks, which breaks the long swings while keeping short autocorrelation, and runs the identical rolling and nested sweep on each resample. Across 250 resamples the first component share averages 0.86 to 0.88 and reaches the observed 0.98 in 2 percent of resamples at a 24 month block, 6 percent at 48 months and 12 percent at 96

Table 3. All stock summaries, on the lower bound $SR_t^L(c)$. start is the Dec 1982 window, peak is the Aug 1999 window, end is Dec 2025. The trend t is a linear trend in the level with a Newey-West standard error at lag 240, read as a diagnostic per the note below. The SVD share is 0.98 on the lower bound, 0.98 on the midpoint and 0.99 on the upper bound.

	Matched $\hat{Z}'_*$	Fixed z
Level $SR_t^L(c = 20.8)$, start / peak / end	6.7 / 12.8 / 5.2	7.2 / 13.8 / 6.3
Ratio $SR_t^L(c = 20.8)/SR_t^L(c = 0.25)$, mean	1.8	2.0
Difference $SR_t^L(c = 20.8) - SR_t^L(c = 0.25)$, mean	3.6	4.3
Newey-West trend t on the level	-1.7	-1.6
SVD first component share	0.98	0.98

months. The observed surface is more nearly one factor than overlapping windows alone tend to produce, and the check is suggestive rather than decisive, since the margin narrows as the resampling block preserves more of the low frequency persistence.

4 Complexity sweep by size group

The sweep in the previous section pools all stocks. This section runs the same sweep separately for the four CKMS size groups, mega, large, small and micro, defined by the 80th, 50th, 20th and 1st NYSE market capitalization percentiles as in Table 1. The 240 month window, the monthly step, the 517 windows, the 13 feature counts and the two ridge conventions are the same as in the all stock sweep, and Ω is the same draw with the smaller models again using its first rows. Each size group yields a surface $SR_t^L(c)$ of 517 windows by 13 feature counts under matched $\hat{Z}'_*$ and under fixed z , the $J = 1$ lower bound as in Section 3.

For the size groups the plug in covariance is not positive semidefinite at the higher feature counts, so the delta method confidence limits are not available and the surfaces hold point bounds only. The departures table at the end of this report records the same limitation for the size groups in the replication. Formal trend inference is not reported by size group, for the reason in the note below. Figure 4 shows the surfaces under both conventions, Figure 5 shows the reduction panels for mega, and Table 4 collects the summaries. Every result in this section holds under both ridge conventions, with the fixed z values in Table 4 tracking the matched ones.

Table 4. Size group summaries on the lower bound $SR_t^L(c)$ under matched $\hat{Z}'_*$, with the fixed z value in parentheses. start is the Dec 1982 window and end is the Dec 2025 window. peak is the window with the highest level, where the level is $SR_t^L(c = 20.8)$. ratio and difference are $SR_t^L(c = 20.8)$ over and minus $SR_t^L(c = 0.25)$, averaged over windows. peak to end is the percent fall in the level from the peak window to the Dec 2025 window. The SVD share is the first component's share of the time variation in the log surface after the time mean is removed at each c . wedge is the largest nonlinear lower bound minus the linear lower bound, averaged over windows.

	Mega	Large	Small	Micro
Level, start	2.8 (3.0)	4.1 (4.9)	4.5 (4.9)	6.4 (7.5)
Level, peak	3.4 (4.3)	4.9 (5.7)	7.3 (8.3)	12.0 (13.6)
Peak window	Mar 2015 (Aug 2012)	Sep 1987	Jan 2001	Jul 2003
Level, end	1.9 (2.9)	2.0 (2.6)	2.7 (4.2)	4.9 (6.7)
Ratio, mean	1.8 (2.2)	1.9 (2.2)	1.6 (2.0)	1.6 (1.9)
Difference, mean	1.2 (1.7)	1.6 (2.3)	1.9 (2.8)	3.0 (4.2)
Level, peak to end fall, pct	43 (31)	59 (54)	63 (50)	60 (51)
SVD first component share	0.74 (0.67)	0.97 (0.94)	0.99 (0.98)	0.99 (0.99)
Wedge, mean	0.4 (1.0)	0.8 (1.5)	0.9 (2.0)	1.6 (2.9)

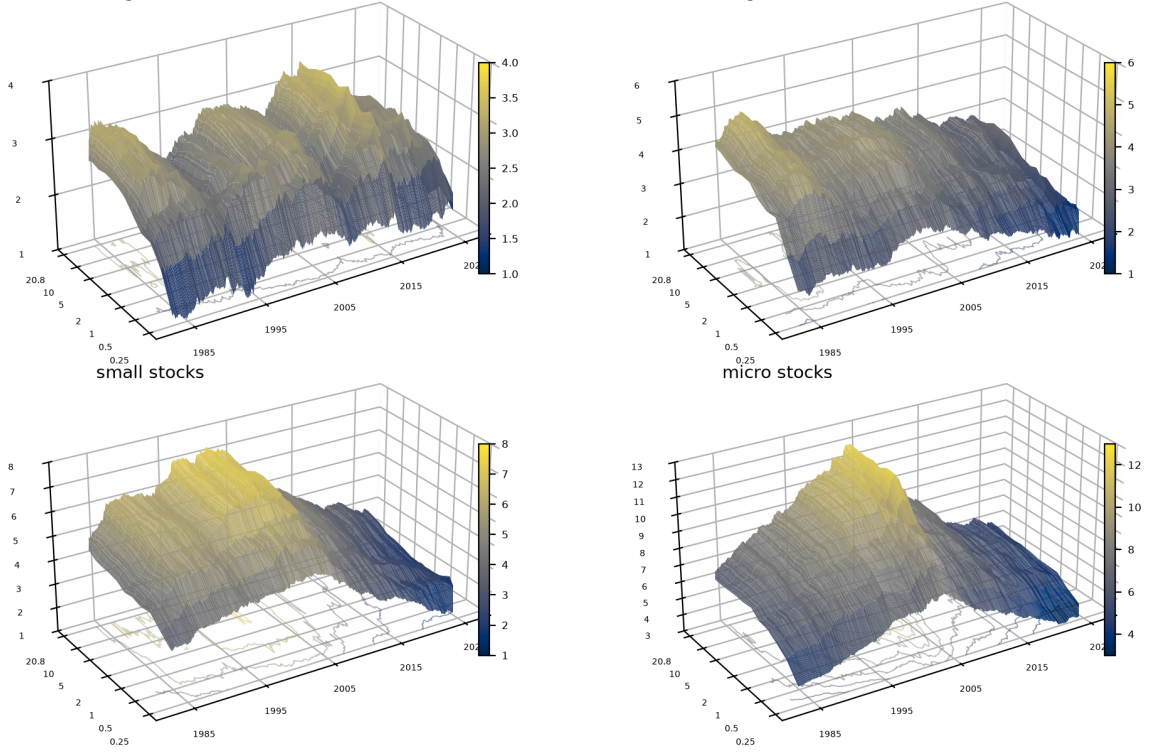
Three regularities hold for large, small and micro. First, the level rises as size decreases in almost every window. At the largest model the estimate is ordered mega below large below small below micro in 95 percent of windows under matched $\hat{Z}'_*$ and in 88 percent under fixed z , and this ordering matches the full sample ordering in CKMS Figure 4. Second, a singular value decomposition of the log surface, after the time mean is removed at each c , places 0.97 or more of the time variation on one component for each of these three groups under both conventions, and that component correlates 0.95 or higher with the largest model level. As in the all stock sweep, the surface for these groups is close to a fixed shape scaled by a time varying level, and the stationary block resampling check in the complexity sweep section puts the micro share of 0.99 above its null at every block length tested. Third, that level traces one rise and fall. The micro level rises from about 6.4 at the December 1982 window to about 12.0 at the July 2003 window and falls to about 4.9 by December 2025. The small level rises from about 4.5 to about 7.3 at the January 2001 window and falls to about 2.7. The large level stays between about 4.1 and 4.9 from the mid 1980s to the mid 2000s and is about 2.0 at the end of the sample. Measured from each group's highest window to the December 2025 window, the matched level falls 59 percent for large, 63 percent for small and 60 percent for micro.

The mega surface differs from the other three on each point above. Its first singular value decomposition component carries 0.74 of the time variation under matched $\hat{Z}'_*$ and 0.67 under fixed z , with a second component between 0.14 and 0.21, so the mega surface is not close to one time varying level. The mega level stays between about 1.9 and 4.3 across the whole sample with several local high points rather than one, and its highest window is March 2015 under matched $\hat{Z}'_*$. From that window to December 2025 the matched mega level falls 43 percent, against 59 percent or more for the other three groups.

The returns to complexity are close in proportional terms across the four groups and larger in absolute terms for the smaller groups. The average ratio $SR_t^L(c = 20.8)/SR_t^L(c = 0.25)$ is between 1.6 and 2.2 for every group under both conventions. The average difference $SR_t^L(c = 20.8) - SR_t^L(c = 0.25)$ rises as size decreases, from about 1.2 Sharpe points for mega to about 3.0 for micro under matched $\hat{Z}'_*$. Neither series shows a clear trend across windows for any group. The all stock level is close to the micro level in Table 2, but the proportional gain is similar across all four size groups, so the all stock frontier shape is not only a restatement of the micro group.

The gap between the largest nonlinear lower bound and the linear lower bound is smallest for

Estimated maximal Sharpe ratio ($J=1$ lower bound) over complexity $c = P / T$ and window end year, by size group, matched $\hat{Z}'_*$



Estimated maximal Sharpe ratio ($J=1$ lower bound) over complexity $c = P / T$ and window end year, by size group, fixed z

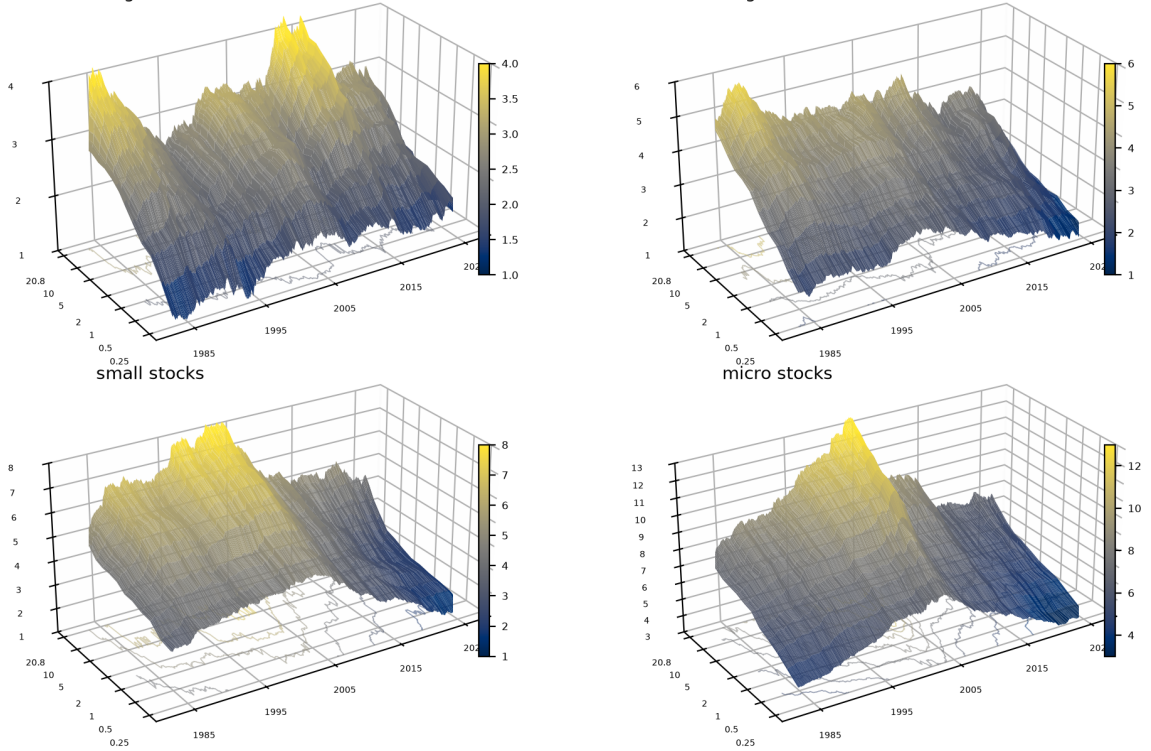


Figure 4. $SR_t^L(c)$, the $J = 1$ lower bound, for the four size groups, under matched $\hat{Z}'_*$ (top) and fixed z (bottom). Each surface is 517 windows by 13 feature counts. The floor axis running left is $c = P/T$ from 0.25 to 20.8, the floor axis running right is the window end year, and the height is the annualized lower bound, with the vertical scale set per size group.

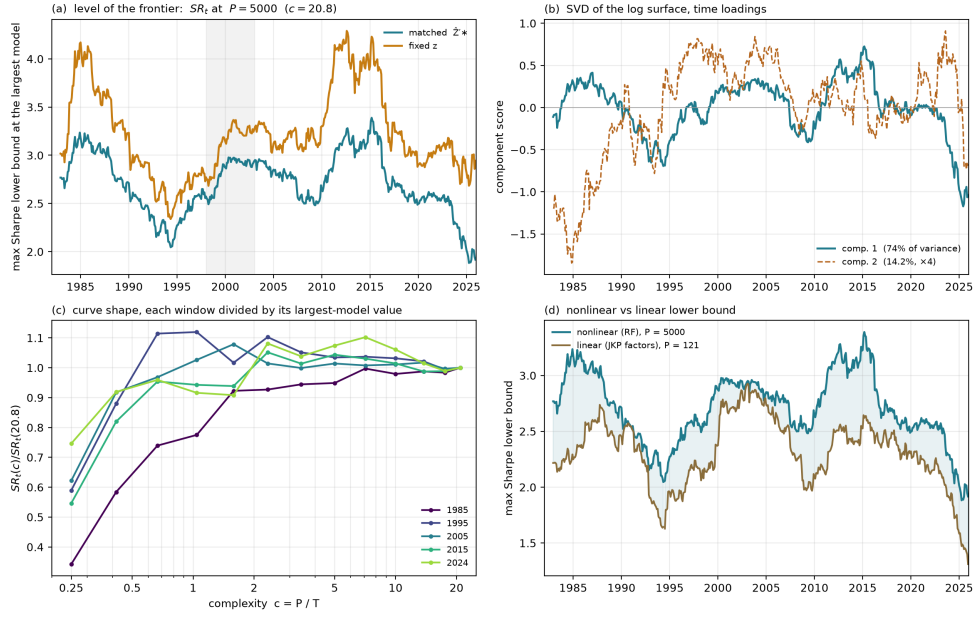


Figure 5. Reduction panels for mega stocks, with the panels defined as in Figure 3. The component score in panel (b) has an arbitrary sign. The large, small and micro panels show a single dominant time component and are in the results directory.

mega and rises as size decreases. Its average under matched $\hat{Z}'_*$ is 0.4 Sharpe points for mega, 0.8 for large, 0.9 for small and 1.6 for micro, and under fixed z it is 1.0, 1.5, 2.0 and 2.9. This gap does not trend for any group.

That gap compares two feature constructions, the 121 linear portfolios and the 5000 nonlinear ones, so it mixes a change in functional form with a change in dimension. A nested check holds the linear span fixed and adds nonlinear directions to it, with managed returns from $[X_t \text{ portfolios}, \text{ReLU}(X_t \Omega'_{1:P})]$ and total dimension $121 + P$, so $P = 0$ is the 121 dimensional linear opportunity set and every increase in P adds nonlinear capacity while keeping the linear span. Adding all 5000 nonlinear directions to the linear span raises the lower bound by about 0.4 Sharpe points for mega, 0.8 for large, 0.9 for small, 1.5 for micro and 1.9 for all stocks under matched $\hat{Z}'_*$, within about 0.1 of the raw gap above in every group, so the raw gap is a close proxy for the incremental value of nonlinear capacity here. In proportional terms the nested increment multiplies the linear span estimate by about 1.2 to 1.3 across the four size groups under matched $\hat{Z}'_*$ and about 1.5 under fixed z , without a clear trend across windows.

The level ends the sample well below its peak in every group, by 43 percent for mega and 59 to 63 percent for the other three. Formal trend inference is not reported by size group. As the note below sets out, 63 years of monthly data hold only about three independent 240 month windows, and each size group holds a smaller cross section than the all stock universe, so the per window point estimates are noisier here than in the all stock sweep.

Important note

The rolling estimates use 240 month windows over 756 months, so $756/240 \approx 3$ non overlapping windows. Stepping one month forward changes 1 of the 240 months, so consecutive estimates are nearly identical and the number of independent windows is about three. The Newey-West standard error at lag 240 in Table 3 treats this overlap. Under it the decline in the level has

$t \approx -1.8$, which does not reach the usual thresholds, and with only about three non overlapping windows a HAC calculation is itself a diagnostic rather than formal inference. Either way, 63 years cannot separate a decline from a long swing.

Each window also covers 240 months, so a value dated December 1999 summarizes the opportunity set over January 1980 through December 1999, not that single year. The rise and fall in the level is a smoothed 20 year object. $T = 240$ is used because it is the CKMS specification. Repeating the all stock sweep at $T = 120$ and $T = 360$ leaves the picture in place, the first component still carries 98 to 99 percent of the time variation and the level still traces one hump, with the hump peak near the mid 2002 window at $T = 360$ and the early 1998 window at $T = 120$.

Where this build differs from CKMS

The table below lists the points where this replication departs from the paper, in the order the steps occur in the pipeline.

Table 5. Departures from CKMS, in pipeline order.

Step	CKMS	This build
Characteristic set	132 by the $< 30\%$ missing rule	121 by the same rule on this data vintage, most likely a vintage or preprocessing difference
Missing return rows	not stated in the paper	dropped, no imputation, 2,627,813 of 2,635,914 eligible stock months kept
Ridge z , return scale	10^{-4} on the paper's scale	the ridge scale is not aligned with the paper's here, so results are shown for fixed $z = 2.5 \times 10^{-7}$ and matched $\hat{Z}'_* = 1.4$, with $z = 3 \times 10^{-8}$ as a one off scale check
Shrinkage $\hat{Z}'_*$, full sample	≈ 1.8	≈ 1.7 at the scale check, ≈ 1.3 at fixed z
Implied ridge ceiling $\sqrt{12L/\hat{Z}'_*}$	≈ 5.5	≈ 5.8 at fixed z
Full sample all stock, $\sqrt{12H}$ as L / U	7.43 / 7.42	7.50 / 7.48 at the scale check, 6.41 / 6.57 at fixed z
Size groups	Figure 4 boxes	within about 10 percent at the scale check, ordering preserved
Standard error band	delta method	a Monte Carlo check indicates this band understates the spread, and the plug in covariance is not positive semidefinite for the size subgroups, so rolling inference uses Newey-West at the window length